Constant Speed Motion: Graphs of Constant Speed Motion

<u>Synopsis</u>: Recall that we describe motion as distance vs. time (position as a function of time) or speed vs. time (speed as a function of time). In constant speed motion, you saw that the distance vs. time graph is a linearly increasing plot with the slope representing the speed, the rate of change of distance. The faster your speed, the steeper the line. Does the slope of distance vs time change with time for the constant speed motion? No. You know from your algebra that in a linear graph, the slope is constant. This is consistent with your experience – you did not speed up or slow down; there was no acceleration.

We must discuss *<u>speed</u>* vs. time for completeness because this is another way we describe motion. However, you should notice that the answer is trivial for a constant-speed motion – it's constant, unchanging, and the same, so it does not vary with time.

<u>Performance Expectations</u>: Recognize that in constant speed motion, the slope of the distance vs. time graph is constant (but not zero slope, which is a flat line), indicating a constant rate of change or speed. The speed vs. time graph is a flat line, showing a zero slope (you should recognize this corresponds to zero acceleration; the speed change rate is zero).

Constant speed motion (acceleration = 0)
Distance vs. Time
Speed vs. Time
Acceleration vs. Time

Sketch motion graphs

In Calculus Notation

Velocity is the instantaneous rate of change of position: $v = dx/dt$. For constant-speed motion, this derivative is itself constant, so the second derivative — the acceleration — is zero:

$dx/dt = \text{constant}$ $\quad a = d^2x/dt^2 = 0$